

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

Герчет Вячеслав

2026-04-03

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
vgerchet@vgerchet:~$ ls /etc/ > file.txt
vgerchet@vgerchet:~$ ls >> file.txt
vgerchet@vgerchet:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
```

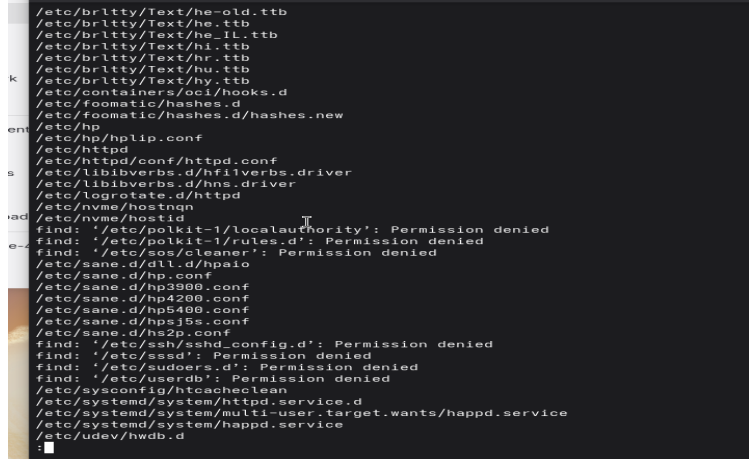
Рисунок 1: Запись в файл

```
vgerchet@vgerchet:~$ grep .conf file.txt > conf.txt
vgerchet@vgerchet:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
makedumpfile.conf.sample
man_db.conf
```

Рисунок 2: Поиск расширения .conf


```
/home/vgerchet/.cache/mesa_shader_cache/66/c95f1f476709ae97d9ab65d13b670a8b453714
/home/vgerchet/.cache/mesa_shader_cache/0e/cea9f92be84b66c9c2c80ccea205510ff87413
/home/vgerchet/.cache/mesa_shader_cache/71/cd651132ebecdb69c3c6b33e265dbf1f978e02
/home/vgerchet/.config/pulse/cookie
/home/vgerchet/.config/mozilla/firefox/Crash Reports/crash_helper_server.log
/home/vgerchet/.config/mozilla/firefox/znv0ri7k.default-release/crashes
/home/vgerchet/.config/mozilla/firefox/znv0ri7k.default-release/compatibility.ini
/home/vgerchet/.config/mozilla/firefox/znv0ri7k.default-release/cookies.sqlite
/home/vgerchet/.config/mozilla/firefox/znv0ri7k.default-release/storage/permanent/chrome
/home/vgerchet/.config/mozilla/firefox/znv0ri7k.default-release/cert9.db
/home/vgerchet/.config/mozilla/firefox/znv0ri7k.default-release/security_state/crlite.fil
ter
/home/vgerchet/.config/mozilla/firefox/znv0ri7k.default-release/content-prefs.sqlite
/home/vgerchet/.config/mozilla/firefox/znv0ri7k.default-release/datareporting/glean/clien
t_id.txt
/home/vgerchet/.config/mozilla/firefox/znv0ri7k.default-release/containers.json
/home/vgerchet/.config/mozilla/firefox/znv0ri7k.default-release/cookies.sqlite-wal
/home/vgerchet/.local/share/flatpak/repo/tmp/cache
/home/vgerchet/.local/share/flatpak/repo/config
/home/vgerchet/.local/share/evolution/addressbook/system/contacts.db
/home/vgerchet/.local/share/evolution/calendar
/home/vgerchet/.local/share/evolution/calendar/system/calendar.ics
/home/vgerchet/.local/share/containers
/home/vgerchet/.local/share/containers/storage/overlay-containers/containers.lock
/home/vgerchet/conf.txt
vgerchet@vgerchet:~$
```

Рисунок 3: Поиск файлов



```
/etc/brltty/Text/he-old.ttb
/etc/brltty/Text/he.ttb
/etc/brltty/Text/he_IL.ttb
/etc/brltty/Text/hl.ttb
/etc/brltty/Text/hr.ttb
/etc/brltty/Text/hu.ttb
/etc/brltty/Text/hy.ttb
/etc/containers/oci/hooks.d
/etc/foomatic/ashes.d
/etc/foomatic/ashes.d/ashes.new
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
/etc/logrotate.d/httpd
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
/etc/sysconfig/htcacheclean
/etc/systemd/system/httpd.service.d
/etc/systemd/system/multi-user.target.wants/happd.service
/etc/systemd/system/happd.service
/etc/udev/hwdb.d
:
```

Рисунок 4: Поиск файлов

```
vgerchet@vgerchet:~$  
vgerchet@vgerchet:~$ find /etc -name "h*" -print | less  
find: '/etc/audit': Permission denied  
find: '/etc/credstore.encrypted': Permission denied  
find: '/etc/credstore': Permission denied  
find: '/etc/cups/ssl': Permission denied  
find: '/etc/firewalld': Permission denied  
find: '/etc/grub.d': Permission denied  
find: '/etc/libvirt': Permission denied  
find: '/etc/lvm/archive': Permission denied  
find: '/etc/lvm/backup': Permission denied  
find: '/etc/lvm/cache': Permission denied  
find: '/etc/lvm/devices': Permission denied  
find: '/etc/nftables': Permission denied  
find: '/etc/openvpn/client': Permission denied  
find: '/etc/openvpn/server': Permission denied  
find: '/etc/pki/rsyslog': Permission denied  
vgerchet@vgerchet:~$  
vgerchet@vgerchet:~$  
vgerchet@vgerchet:~$ find ~ -name "log*" > logfile &  
[1] 8102  
vgerchet@vgerchet:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
vgerchet@vgerchet:~$ rm logfile  
vgerchet@vgerchet:~$
```

Рисунок 5: Фоновый запуск процесса

```
vgerchet@vgerchet:~$  
vgerchet@vgerchet:~$ gedit &  
[1] 8202  
vgerchet@vgerchet:~$ ps | grep gedit  
    8202 pts/0    00:00:00 gedit  
vgerchet@vgerchet:~$ kill  
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigsp  
ec]  
vgerchet@vgerchet:~$ kill 8202  
[1]+  Terminated                  gedit  
vgerchet@vgerchet:~$  
vgerchet@vgerchet:~$
```

Рисунок 6: Фоновый запуск и завершение процесса



```
vgerchet@vgerchet:~ — man df
DE(1)                                User Commands                                DE(1)

NAME
  df - report file system space usage

SYNOPSIS
  df [OPTION]... [FILE]...

DESCRIPTION
  This manual page documents the GNU version of df.  df displays the amount of
  space available on the file system containing each file name argument.  If no
  file name is given, the space available on all currently mounted file systems is
  shown.  Space is shown in 1K blocks by default, unless the environment variable
  POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

  If an argument is the absolute file name of a device node containing a mounted
  file system, df shows the space available on that file system rather than on the
  file system containing the device node.  This version of df cannot show the space
  available on unmounted file systems, because on most kinds of systems doing so
  requires non-portable intimate knowledge of file system structures.

OPTIONS
  Show information about the file system on which each FILE resides, or all file
  systems by default.

  Mandatory arguments to long options are mandatory for short options too.

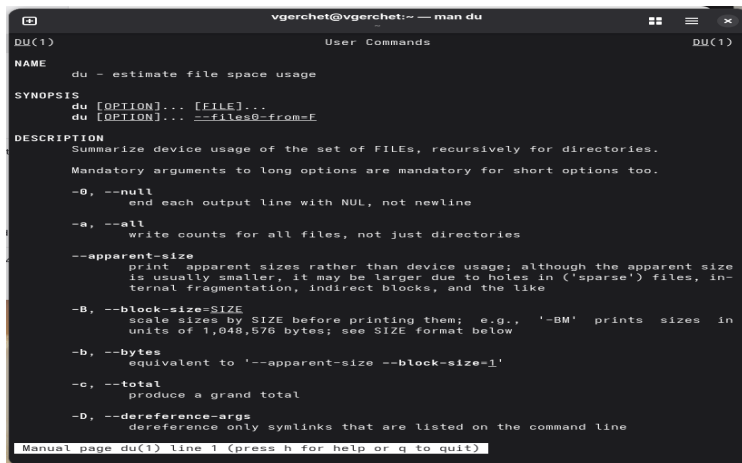
  -a, --all
      include pseudo, duplicate, inaccessible file systems

  -B, --block-size=SIZE
      scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in
      units of 1,048,576 bytes; see SIZE format below

  --direct
      show statistics for a file instead of mount point

  -h, --human-readable
      Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df



```
vgerchet@vgerchet:~ — man du
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size
        is usually smaller, it may be larger due to holes in ('sparse') files, in-
        ternal fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in
        units of 1,048,576 bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

    -c, --total
        produce a grand total

    -D, --dereference-args
        dereference only symlinks that are listed on the command line

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
vgerchet@vgerchet:~$ df
vgerchet@vgerchet:~$ df
Filesystem      1K-blocks      Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 101167040 53666272  66% /
devtmpfs        4011292      0    4011292   0% /dev
tmpfs           4035664       92   4035572   1% /dev/shm
tmpfs           1614268     1944   1612324   1% /run
tmpfs            1024         0        1024   0% /run/credentials/systemd-journald.service
/dev/nvme0n1p3 155186176 101167040 53666272  66% /home
tmpfs           4035668     5668   4030000   1% /tmp
/dev/nvme0n1p2  1992552     518800   1352512  28% /boot
/dev/loop2       49280       49280         0 100% /var/lib/napd/snap/snapd/25935
/dev/loop0       115712     115712         0 100% /var/lib/napd/snap/hugo/25676
/dev/loop1       68480       68480         0 100% /var/lib/napd/snap/core24/1349
tmpfs            1024         0        1024   0% /run/credentials/systemd-resolved.service
tmpfs           807132       192    806940   1% /run/user/1083
/dev/sr0         2677920    2677920         0 100% /run/media/vgerchet/Fedora-WS-Live-43
vgerchet@vgerchet:~$
```

Рисунок 9: Справка по команде du

```
0      ./local/share/evolution/memos/trash
0      ./local/share/evolution/memos
0      ./local/share/evolution/tasks/trash
4      ./local/share/evolution/tasks/system
4      ./local/share/evolution/tasks
100    ./local/share/evolution
76     ./local/share/gvfs-metadata
8      ./local/share/keyrings
208    ./local/share/nautilus/tags
0      ./local/share/nautilus/scripts
208    ./local/share/nautilus
0      ./local/share/containers/storage/overlay-containers
0      ./local/share/containers/storage/libpod
0      ./local/share/containers/storage/volumes
0      ./local/share/containers/storage/overlay/l
4      ./local/share/containers/storage/overlay
0      ./local/share/containers/storage/overlay-images
0      ./local/share/containers/storage/networks
0      ./local/share/containers/storage/overlay-layers
124    ./local/share/containers/storage
124    ./local/share/containers
0      ./local/share/org.gnome.Ptyxis/palettes
0      ./local/share/org.gnome.Ptyxis
528    ./local/share
532    ./local
239916 .
vgerchet@vgerchet:~$
```

Рисунок 10: Запуск команды du


```
/home/vgerchet/.local/share/evolution/addressbook/trash
/home/vgerchet/.local/share/evolution/addressbook/system
/home/vgerchet/.local/share/evolution/addressbook/system/photos
/home/vgerchet/.local/share/evolution/calendar
/home/vgerchet/.local/share/evolution/calendar/trash
/home/vgerchet/.local/share/evolution/calendar/system
/home/vgerchet/.local/share/evolution/mail
/home/vgerchet/.local/share/evolution/mail/trash
/home/vgerchet/.local/share/evolution/memos
/home/vgerchet/.local/share/evolution/memos/trash
/home/vgerchet/.local/share/evolution/tasks
/home/vgerchet/.local/share/evolution/tasks/trash
/home/vgerchet/.local/share/evolution/tasks/system
/home/vgerchet/.local/share/gvfs-metadata
/home/vgerchet/.local/share/keyrings
/home/vgerchet/.local/share/nautilus
/home/vgerchet/.local/share/nautilus/tags
/home/vgerchet/.local/share/nautilus/scripts
/home/vgerchet/.local/share/containers
/home/vgerchet/.local/share/containers/storage
/home/vgerchet/.local/share/containers/storage/overlay-containers
/home/vgerchet/.local/share/containers/storage/libpod
/home/vgerchet/.local/share/containers/storage/volumes
/home/vgerchet/.local/share/containers/storage/overlay
/home/vgerchet/.local/share/containers/storage/overlay/l
/home/vgerchet/.local/share/containers/storage/overlay-images
/home/vgerchet/.local/share/containers/storage/networks
/home/vgerchet/.local/share/containers/storage/overlay-layers
/home/vgerchet/.local/share/org.gnome.Ptyxis
/home/vgerchet/.local/share/org.gnome.Ptyxis/palettes
vgerchet@vgerchet:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.